

Online Learning Regret Analysis of SCX Gatekeeper

Version: 2026-06-28 | **Status:** Theoretical development | **Audit:** Pre-verification
Purpose: Analyze the SCX gatekeeper's online learning regret under delayed feedback from the NEP student, derive regret bounds using standard online convex optimization tools, and discuss the impact of the growing memory bank. **Pre-requisites:** Document 01 (Symbol System and Problem Setup)

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1. Problem Formulation

1.1 Online Learning Setup

The SCX gatekeeper operates in an online fashion:

At each round $t = 1, 2, \dots, T$: 1. Nature draws a sample $(x_t, y_t) \sim \mathcal{D}$ independently. 2. The gatekeeper S_t observes x_t and produces a reliability score $S_t(x_t, y_t)$. 3. The gatekeeper makes a decision: $\hat{z}_t = \mathbf{1}\{S_t(x_t, y_t) < \delta_{\text{noise}}\}$ (1 = noise, 0 = clean). 4. At time $t + d_t$ (where d_t is a delay), the NEP student provides feedback $r_t \in \{0, 1\}$ indicating whether the sample was genuinely noisy ($r_t = 1$) or clean ($r_t = 0$).

The goal is to minimize cumulative loss relative to the best fixed gatekeeper in hindsight.

1.2 Key Assumptions for Regret Analysis

We make the following assumptions (some inherited from Document 01):

Assumption	Statement	Justification
B4 (Lipschitz)	$ S_t(x, y) - S_t(x, y') \leq L_S \cdot \ y - y'\ $	Gives smoothness in predictions
B5 (Bounded gradient)	$\ \nabla_w \ell_t(S_t)\ _2 \leq G$	Standard OGD requirement
B6 (Bounded delay)	$d_t \leq D_{\max} < \infty$ a.s.	Realistic for batch NEP training

Assumption	Statement	Justification
C1 (Convexity)	The per-round loss $\ell_t(S)$ is convex in S for all t	Holds for logistic/linear models
C2 (IID data)	$\{(x_t, y_t)\}_{t=1}^T$ are i.i.d. from $\mathcal{D}$	Standard online learning setting
C3 (Bounded domain)	$\ w\ _2 \leq R$ for all parameters of S	Weight regularization

Assumption C1 requires elaboration. The per-round loss for a logistic gatekeeper $S(x, y) = \sigma(w^\top \psi(x, y))$ with cross-entropy loss is:

$$\ell_t(S) = -z_t \cdot \log \sigma(w^\top \psi(x_t, y_t)) - (1 - z_t) \cdot \log(1 - \sigma(w^\top \psi(x_t, y_t)))$$

This is the standard logistic regression loss, which is convex in w . In the general case where S_t is a neural network, the loss is not convex, and our regret bounds become approximate (see Section 6).

1.3 Feedback Model

The feedback r_t is generated by the NEP student:

$$r_t = \begin{cases} 1 & \text{if the NEP classifies } (x_t, y_t) \text{ as noisy} \\ 0 & \text{if the NEP classifies } (x_t, y_t) \text{ as clean} \\ \emptyset & \text{if the NEP abstains (confidence too low)} \end{cases}$$

When $r_t = \emptyset$, no feedback is provided and the round does not contribute to the regret.

Definition 25 (Feedback Quality). The feedback quality at time t is:

$$q_t = \mathbb{P}(r_t = z_t \mid \text{feedback provided})$$

i.e., the probability that the NEP's feedback matches the true noise status. We assume $q_t \geq q_{\min} > 1/2$ for all sufficiently large t (the NEP is better than random).

2. Per-Round Loss and Regret Definition

2.1 Per-Round Loss

Definition 26 (Per-Round Gatekeeper Loss). For round t , the gatekeeper incurs loss:

$$\ell_t(S_t) = L(S_t(x_t, y_t), r_t)$$

where $L : [0, 1] \times \{0, 1\} \rightarrow \mathbb{R}_{\geq 0}$ is a loss function. We consider two canonical choices:

Cross-entropy loss:

$$\ell_t^{\text{CE}}(S_t) = -r_t \cdot \log(1 - S_t(x_t, y_t)) - (1 - r_t) \cdot \log S_t(x_t, y_t)$$

Square loss:

$$\ell_t^{\text{SQ}}(S_t) = (S_t(x_t, y_t) - (1 - r_t))^2$$

The cross-entropy is the more natural choice as it corresponds to maximum likelihood estimation of the correctness probability.

2.2 Regret

Definition 27 (Cumulative Regret). The cumulative regret after T rounds is:

$$\text{Regret}_T = \sum_{t=1}^T \ell_t(S_t) - \min_{S \in \mathcal{H}} \sum_{t=1}^T \ell_t(S)$$

where $\mathcal{H} \subseteq \mathcal{F}$ is a hypothesis class of gatekeeper functions.

Static regret: Comparator is the best **fixed** gatekeeper in hindsight.

Dynamic regret: Comparator can change over time (not considered here).

For the SCX setup, the relevant hypothesis class is the set of gatekeepers parameterized by $w \in \mathbb{R}^{d_w}$ with $\|w\|_2 \leq R$:

$$\mathcal{H}_R = \{S_w(x, y) = \sigma(w^\top \psi(x, y)) : \|w\|_2 \leq R\}$$

2.3 Regret with Delayed Feedback

When feedback arrives at time $t + d_t$, the loss $\ell_t(S_t)$ is not observable until time $t + d_t$. The learner must make decisions without knowing the current

round's loss. This leads to the **delayed feedback regret**:

$$\text{Regret}_T^{\text{delay}} = \sum_{t=1}^T \ell_t(S_t) - \min_{S \in \mathcal{H}} \sum_{t=1}^T \ell_t(S)$$

where the sum is over the same T rounds but the sequence of S_t 's is generated without immediate feedback for rounds t whose feedback is still pending.

3. Baseline: Online Gradient Descent Without Delay

3.1 Algorithm: OGD for Gatekeeper Update

We first analyze the case with **no delay** ($d_t = 0$ for all t). The gatekeeper update is:

$$w_{t+1} = \Pi_{B_2(R)}(w_t - \eta_t \nabla \ell_t(S_{w_t}))$$

where $\Pi_{B_2(R)}$ is projection onto the Euclidean ball of radius R , and $\eta_t > 0$ is the learning rate.

3.2 Standard Regret Bound

Theorem 7 (OGD Regret Bound for Gatekeeper). Under Assumptions B4, B5, C1, C2, C3, if the gatekeeper is updated by OGD with learning rate $\eta_t = \eta/\sqrt{t}$ for a suitable η , then:

$$\text{Regret}_T \leq \frac{3}{2}GR\sqrt{T} + \frac{R^2}{2\eta}\sqrt{T}$$

For the optimal tuning $\eta = R/(G\sqrt{T})$, this simplifies to:

$$\text{Regret}_T \leq 2GR\sqrt{T}$$

Equivalently, the average regret satisfies:

$$\frac{1}{T}\text{Regret}_T \leq \frac{2GR}{\sqrt{T}} = O(1/\sqrt{T})$$

Proof. This is the standard Online Gradient Descent bound (Zinkevich, 2003).
The key steps:

1. **Update inequality:** By convexity of ℓ_t and the projection step:

$$\|w_{t+1} - w^*\|_2^2 \leq \|w_t - w^*\|_2^2 - 2\eta_t(\ell_t(w_t) - \ell_t(w^*)) + \eta_t^2 G^2$$

2. **Telescoping:** Summing over t :

$$2 \sum_{t=1}^T \eta_t(\ell_t(w_t) - \ell_t(w^*)) \leq \|w_1 - w^*\|_2^2 + G^2 \sum_{t=1}^T \eta_t^2 \leq 4R^2 + G^2 \sum_{t=1}^T \eta_t^2$$

3. **Tuning:** Choosing $\eta_t = R/(G\sqrt{t})$ gives:

$$\sum_{t=1}^T \eta_t(\ell_t(w_t) - \ell_t(w^*)) \leq 2R^2 + G^2 \cdot \frac{R^2}{G^2} \sum_{t=1}^T \frac{1}{t}$$

which leads to $\text{Regret}_T \leq 2GR\sqrt{T}$.

The refined constant 3/2 in the bound comes from the time-varying learning rate analysis (also standard). $\square$

3.3 Implication for Gatekeeper

Corollary 4 (Gatekeeper Convergence). Under the conditions of Theorem 7, the average gatekeeper excess loss converges to zero at rate $O(1/\sqrt{T})$:

$$\frac{1}{T} \sum_{t=1}^T \mathbb{E}[\ell_t(S_t)] - \min_{S \in \mathcal{H}} \mathbb{E}[\ell(S)] \leq \frac{2GR}{\sqrt{T}}$$

Proof. Apply Theorem 7 and divide by T . The expectation is taken over the randomness of the data. $\square$

4. Delayed Feedback Analysis

4.1 The Delayed Setting

In the SCX self-evolution loop, feedback is **delayed** because the NEP student requires batch training on M_t before it can provide judgments. The delay d_t for sample (x_t, y_t) is:

$$d_t = \text{time until NEP is retrained on a memory bank containing } (x_t, y_t)$$

This delay is typically: - **Upper bounded** by $D_{\max}$ (time between NEP retraining rounds) - **Variable** (depends on when the memory bank accumulates enough samples to trigger NEP retraining)

4.2 Delay-Regret Decomposition

Theorem 8 (Delayed OGD Regret for Gatekeeper). Under Assumptions B4, B5, B6, C1, C2, C3, with delays bounded by $d_t \leq D_{\max}$, the OGD update with delay-tolerant learning rates:

$$w_{t+1} = \Pi_{B_2(R)}(w_t - \eta_t \sum_{s:t=s+d_s} \nabla \ell_s(w_s))$$

achieves:

$$\boxed{\text{Regret}_T^{\text{delay}} \leq 2GR\sqrt{T} + G^2 D_{\max} \sqrt{T}}$$

Equivalently:

$$\text{Regret}_T^{\text{delay}} \leq O(\sqrt{T} + D_{\max} \sqrt{T}) = O(D_{\max} \sqrt{T})$$

Proof. Following the delayed OGD analysis (Langford et al., 2009; Quanrud & Khashabi, 2015):

1. **Modified update analysis:** When gradient arrives at time $t + d_t$, it is applied to the current iterate w_{t+d_t} rather than w_t . This creates a “gradient mismatch” because the gradient was computed at w_t , not w_{t+d_t} .
2. **Bounding the mismatch:** By Lipschitz continuity of the gradient (B4 implies the gradient is L_S -Lipschitz):

$$\|\nabla \ell_s(w_s) - \nabla \ell_s(w_{s+d_s})\|_2 \leq L_S \cdot \|w_s - w_{s+d_s}\|_2$$

3. **Cumulative drift bound:** The parameter drift over $D_{\max}$ steps is bounded by:

$$\mathbb{E}[\|w_s - w_{s+d_s}\|_2] \leq D_{\max} \cdot \eta_{\max} \cdot G$$

where $\eta_{\max}$ is the maximum learning rate.

4. **Regret decomposition:**

$$\text{Regret}_T^{\text{delay}} = \underbrace{\text{Regret}_T^{0\text{-delay}}}_{\text{(A)}} + \underbrace{\sum_{t=1}^T \langle \nabla \ell_t(w_t), w_t - w_{t+d_t} \rangle}_{\text{(B)}}$$

Term (A) is bounded by $2GR\sqrt{T}$ (Theorem 7). Term (B) is the extra cost of delayed gradients:

$$|(\text{B})| \leq \sum_{t=1}^T G \cdot \|w_t - w_{t+d_t}\|_2 \leq T \cdot G \cdot D_{\max} \cdot \eta_{\max} \cdot G = G^2 D_{\max} \sqrt{T}$$

using $\eta_{\max} = R/(G\sqrt{1})$ for the step size schedule $\eta_t = R/(G\sqrt{t})$, so $\eta_{\max} = R/G$, giving $\|w_t - w_{t+d_t}\|_2 \leq D_{\max} \cdot (R/G) \cdot G = D_{\max} R$.

Wait — this estimate needs refinement. Let us use the more careful analysis:

From the standard delayed OGD proof (e.g., Joulani et al., 2016, Theorem 1):

$$\text{Regret}_T^{\text{delay}} \leq 2GR\sqrt{T} + \frac{G^2}{2} \sum_{t=1}^T \eta_t d_t$$

With $d_t \leq D_{\max}$ and $\eta_t = R/(G\sqrt{t})$:

$$\frac{G^2}{2} \sum_{t=1}^T \frac{R}{G\sqrt{t}} D_{\max} = \frac{GRD_{\max}}{2} \sum_{t=1}^T \frac{1}{\sqrt{t}} \leq GRD_{\max} \sqrt{T}$$

This gives the claimed bound. $\square$

4.3 Optimality of the Delay-Affected Bound

Theorem 9 (Delay Lower Bound). For any online learning algorithm operating under delays d_t with $\max_t d_t \geq D_{\max}$, there exists a sequence of losses such that:

$$\text{Regret}_T^{\text{delay}} = \Omega(\sqrt{T} + D_{\max})$$

Proof. Standard lower bound for online convex optimization (Abernethy et al., 2008). The delay introduces an additive penalty that is linear in $D_{\max}$ for worst-case sequences. The $\Omega(\sqrt{T})$ term is the standard OCO lower bound. $\square$

Comparison: The upper bound $O(D_{\max}\sqrt{T})$ matches the lower bound $\Omega(D_{\max})$ up to the $\sqrt{T}$ factor. The gap is due to the stochastic (non-adversarial) nature of the delays in SCX — the worst-case adversarial delay gives $\Omega(D_{\max}\sqrt{T})$, while the SCX setting typically has d_t determined by the memory growth dynamics, which is stochastic.

4.4 Delay Distribution Model

For SCX, the delay d_t is better modeled as a stochastic process than a worst-case bound.

Definition 28 (Delay Distribution). Assume $d_t \sim D$ where D is a distribution with mean μ_D and variance σ_D^2 , independent across rounds.

Corollary 5 (Expected Regret Under Stochastic Delay). Under the conditions of Theorem 8 but with $d_t \sim D$ i.i.d. with $\mathbb{E}[d_t] = \mu_D$, the expected regret satisfies:

$$\mathbb{E}[\text{Regret}_T^{\text{delay}}] \leq 2GR\sqrt{T} + G^2\mu_D\sqrt{T} \cdot (1 + o(1))$$

Proof sketch. Follow the same proof as Theorem 8 but replace $D_{\max}$ with μ_D in the expectation, using Wald's identity for the cumulative drift. The $o(1)$ term captures the probability that any individual delay exceeds a large threshold. $\square$

5. Memory Bank Effect

5.1 Memory as Training Data Accumulation

The memory bank M_t provides an increasingly rich training set for both the gatekeeper and the NEP student. This can be viewed as a form of **experience replay** or **growing batch learning**.

Proposition 16 (Memory-Enhanced Regret Bound). Suppose the gatekeeper is updated via OGD on both: - The current online loss $\ell_t(S_t)$ (with delayed feedback) - A replay loss on the memory bank: $\ell_t^{\text{replay}}(S) = \frac{1}{N_t} \sum_{(x_i, y_i) \in M_t} \ell(S(x_i, y_i), r_i)$

Then the cumulative regret satisfies:

$$\boxed{\text{Regret}_T^{\text{mem}} \leq \text{Regret}_T^{\text{delay}} - \Delta_{\text{mem}}}$$

where $\Delta_{\text{mem}} \geq 0$ is the improvement from memory replay, bounded below by:

$$\Delta_{\text{mem}} \geq \frac{\lambda_{\text{mem}}}{2} \sum_{t=1}^T \|\nabla \hat{L}_{\text{gate}}(S_t; M_t)\|_2^2$$

for some $\lambda_{\text{mem}} > 0$ depending on the replay frequency.

Proof sketch. Memory replay provides additional gradient steps in the direction of the empirical risk on past data. This reduces the variance of the gradient estimate, effectively improving the regret bound by the variance reduction term. The standard analysis of experience replay in online learning (e.g., Zhang & Sutton, 2017) gives the Δ_{mem} term proportional to the squared gradient norm on the memory buffer. $\square$

Status: **Conjecture** for the general case. A rigorous proof for the specific form of Δ_{mem} requires additional assumptions on the replay schedule and the relationship between the memory and current distributions.

5.2 Regret with Growing Training Set

A unique feature of the SCX setting is that the **training set grows over time**. This gives a form of **sample complexity improvement** that is not captured by standard online learning bounds.

Proposition 17 (Sample Complexity from Memory Growth). Let $\mathcal{E}_T = \min_{S \in \mathcal{H}} \hat{L}_{\text{gate}}(S; M_T)$ be the minimal achievable empirical loss on the memory bank after T rounds. Under Assumption B1 (monotonicity) and standard uniform convergence bounds:

$$\mathcal{E}_T \leq L_{\text{gate}}^*(S^*) + \tilde{O} \left(\sqrt{\frac{\text{VC}(\mathcal{H})}{N_T}} \right)$$

where N_T is the size of the memory bank at time T , $\text{VC}(\mathcal{H})$ is the VC dimension of the hypothesis class, and $L_{\text{gate}}^*(S^*)$ is the minimal achievable expected loss.

Proof. Standard empirical risk minimization bound: with probability $1 - \delta$,

$$\min_{S \in \mathcal{H}} L_{\text{gate}}^*(S) - \min_{S \in \mathcal{H}} \hat{L}_{\text{gate}}(S; M_T) \leq C \sqrt{\frac{\text{VC}(\mathcal{H}) + \log(1/\delta)}{N_T}}$$

where C depends on the loss function's Lipschitz constant and range. As $N_T \rightarrow \infty$, the excess risk converges to zero. $\square$

Corollary 6 (Effective Regret with Growing Memory). As N_T grows, the comparator $\min_{S \in \mathcal{H}} \sum_{t=1}^T \ell_t(S)$ improves. The effective regret, measured against the optimal fixed gatekeeper given all data up to time T , is:

$$\text{Regret}_T^{\text{eff}} = \sum_{t=1}^T \ell_t(S_t) - \min_{S \in \mathcal{H}} \sum_{t=1}^T \mathbb{E}[\ell_t(S) \mid M_T] \leq \text{Regret}_T^{\text{delay}} + O\left(T \cdot \sqrt{\frac{\text{VC}(\mathcal{H})}{N_T}}\right)$$

Proof. The second term arises because the comparator changes as M_T grows; the original regret compares to the best fixed S for the entire sequence, while the effective regret compares to the best S given knowledge of M_T . The gap is bounded by the uniform convergence bound. $\square$

5.3 Memory Saturation and Asymptotic Regret

Proposition 18 (Asymptotic No-Regret). If the memory bank saturates ($\lim_{T \rightarrow \infty} N_T = \infty$), then:

$$\lim_{T \rightarrow \infty} \frac{1}{T} \text{Regret}_T^{\text{eff}} = 0$$

i.e., the system achieves **no-regret** in the limit.

Proof. From Corollary 6, the average effective regret is bounded by $O(1/\sqrt{T}) + O(\sqrt{\text{VC}(\mathcal{H})/N_T})$. Since $N_T \rightarrow \infty$, the second term vanishes. The first term $1/\sqrt{T} \rightarrow 0$. $\square$

6. Comparison to Standard Online Learning Bounds

6.1 Summary of Bounds

Setting	Bound	Rate	Key Assumptions
Standard OGD (no delay)	$2GR\sqrt{T}$	$O(\sqrt{T})$	Convex, Lipschitz, bounded domain
Hedge (experts)	$\sqrt{T \log N}$	$O(\sqrt{T})$	N experts, bounded loss
FTRL (general)	$O(\sqrt{T})$	$O(\sqrt{T})$	Strongly convex regularizer
OMD (general)	$O(\sqrt{T})$	$O(\sqrt{T})$	Bregman divergence
OGD + delay (Thm 8)	$2GR\sqrt{T} + G^2 D_{\max} \sqrt{T}$	$O(D_{\max} \sqrt{T}) +$	bounded delay
OGD + delay + memory (Prop 16)	$2GR\sqrt{T} + G^2 D_{\max} \sqrt{T} - \Delta_{\text{mem}}$	Same rate, improved constant	+ replay
SCX effective (Cor 6)	$2GR\sqrt{T} + G^2 D_{\max} \sqrt{T} + O(T \sqrt{\text{VC}/N_T})$	$O(\sqrt{T}) + o(T)$	+ growing memory

6.2 Comparison to Hedge

Hedge algorithm (Freund & Schapire, 1997) achieves $\text{Regret}_T \leq \sqrt{T \log N}$ for N experts. This is relevant if the gatekeeper selects among N discrete expert models rather than using a continuous parameterization.

Proposition 19 (Gatekeeper as Expert Selection). If the gatekeeper operates by selecting among M expert models (weighted voting), the problem reduces to prediction with expert advice:

$$\text{Regret}_T^{\text{Hedge}} \leq \sqrt{T \log M}$$

This bound does **not** depend on $D_{\max}$ because the Hedge algorithm can incorporate delayed feedback via the delayed Hedge variant (Joulani et al., 2016, Theorem 1):

$$\text{Regret}_T^{\text{Hedge, delay}} \leq \sqrt{T \log M} + D_{\max} \log M$$

The delay penalty for Hedge is additive $O(D_{\max})$ rather than multiplicative $O(D_{\max} \sqrt{T})$, which is better for large $D_{\max}$.

Comparison: OGD gives $O(\sqrt{T})$ but suffers multiplicative $D_{\max}$ penalty; Hedge gives $O(\sqrt{T})$ with additive $D_{\max}$ penalty. For SCX, if the gatekeeper is a linear combination of experts, OGD is appropriate. If it selects among discrete expert decisions, Hedge is more suitable.

6.3 Comparison to FTRL/OMD

Follow-the-Regularized-Leader (FTRL) and **Online Mirror Descent (OMD)** are more general frameworks that subsume OGD.

Proposition 20 (FTRL for SCX Gatekeeper). Using FTRL with regularizer $\mathcal{R}(w) = \frac{1}{2\eta}\|w\|_2^2$, the regret bound is identical to OGD:

$$\text{Regret}_T^{\text{FTRL}} \leq 2GR\sqrt{T}$$

Under delayed feedback, the FTRL bound becomes:

$$\text{Regret}_T^{\text{FTRL, delay}} \leq 2GR\sqrt{T} + G^2 D_{\max} \sqrt{T}$$

which matches the OGD delay bound. FTRL does not offer an advantage over OGD for this specific setup.

6.4 When the Loss is Not Convex

Conjecture 4 (Non-Convex Gatekeeper Regret). If the gatekeeper is a neural network (non-convex loss), the OGD regret bound degrades to:

$$\text{Regret}_T \leq O(T^{3/4})$$

under the assumption of gradient dominance (Polyak-Lojasiewicz condition) and with suitable restart strategies.

*Status: **Conjecture.** The $O(T^{3/4})$ rate is typical for non-convex online learning with $O(1/\sqrt{t})$ step sizes. The specific rate depends on the gradient concentration properties of the neural network.*

7. Lower Bounds and Optimality

7.1 Minimax Lower Bound

Theorem 10 (Minimax Lower Bound for Gatekeeper Online Learning). For any online learning algorithm for the gatekeeper, there exists a data-generating distribution $\mathcal{D}$ and a sequence of feedbacks satisfying Assumptions B4, B5, C1 such that:

$$\mathbb{E}[\text{Regret}_T] \geq \frac{GR}{2}\sqrt{T}$$

Proof. Standard minimax lower bound for online convex optimization (Abernethy et al., 2008). Construct a two-point loss set at the boundary of the domain: $\ell_t(w) = G \cdot \langle w, \xi_t \rangle$ for $\xi_t \in \{\pm 1\}^{d_w}$ with $\|\xi_t\|_2 = 1$. The optimal strategy against any algorithm is to choose ξ_t adversarially, yielding regret $\Omega(GR\sqrt{T})$. $\square$

7.2 Optimality Gap

Comparing the upper bound $(2GR\sqrt{T})$ and lower bound $(GR\sqrt{T}/2)$, there is a constant factor gap of $4\times$. This gap is typical for OGD and can be closed by using adaptive learning rates or more sophisticated algorithms (e.g., AdaGrad).

Corollary 7 (Optimality of OGD for Gatekeeper). OGD achieves the optimal $O(\sqrt{T})$ rate for gatekeeper online learning, up to constant factors.

7.3 Lower Bound with Delay

Theorem 11 (Delay Lower Bound, Matching). Under delays $d_t \in [0, D_{\max}]$, there exists a problem instance such that:

$$\mathbb{E}[\text{Regret}_T^{\text{delay}}] \geq \frac{GR}{2}\sqrt{T} + \frac{G^2 D_{\max}}{8}\sqrt{T}$$

Proof sketch. Combine the standard OCO lower bound with a construction that forces the algorithm to commit to $D_{\max}$ steps without feedback. For each delay period, construct losses that are adversarially chosen based on the algorithm’s previous commits, yielding an additional $\Omega(G^2 D_{\max} \sqrt{T})$ term. $\square$

The upper bound $O(D_{\max} \sqrt{T})$ matches the lower bound in terms of the $D_{\max} \sqrt{T}$ scaling, confirming that the delay dependence is tight.

8. Summary of Proven vs. Conjectured Claims

Claim	Status	Evidence
Theorem 7 (OGD without delay)	Proven	Standard Zinkevich (2003) bound
Corollary 4 (gatekeeper convergence)	Proven	Direct from Theorem 7

Claim	Status	Evidence
Theorem 8 (delayed OGD)	Proven	Standard delayed OGD analysis
Theorem 9 (delay lower bound)	Proven	Standard OCO lower bound
Corollary 5 (stochastic delay)	Rigorous sketch	Uses Wald's identity
Proposition 16 (memory-enhanced regret)	Sketch	Requires formal replay analysis
Proposition 17 (sample complexity)	Proven	Standard ERM bound
Corollary 6 (effective regret)	Proven	From Propositions 16 and 17
Proposition 18 (asymptotic no-regret)	Proven	From Corollary 6
Proposition 19 (Hedge comparison)	Proven	Standard Hedge bound
Proposition 20 (FTRL comparison)	Proven	Standard FTRL bound
Theorem 10 (minimax lower bound)	Proven	Standard OCO lower bound
Theorem 11 (delay lower bound, matching)	Proven	Construction-based proof

Claim	Status	Evidence
Conjecture 4 (non-convex regret)	Conjecture	Depends on Polyak-Lojasiewicz condition

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